

# Music Recommendation via SVD, Feature Projection, and Dimensionality Reduction

August 2026



## Abstract

Music is something everyone can feel. Pinning that feeling to a universal standard has proved difficult, so we classify music by genre instead. This project asks a different question: if we classify by measurable audio features — valence, energy, tempo — does the resulting organisation reproduce the genre taxonomy, or replace it?

We build a music recommendation system on a singular value decomposition implemented from first principles, projecting songs into a reduced latent space and ranking by cosine similarity within it. Two design decisions receive particular attention. We use one-sided Jacobi rather than the textbook eigendecomposition of  $A^T A$ , which squares the condition number and destroys the trailing singular values of a correlated feature matrix: on representative data the two differ by eight orders of magnitude in relative accuracy ( $3.27e-11$  against  $5.13e-03$ ). And we select the number of latent dimensions by explicit criteria rather than by convention.

We then evaluate the system, which is the part most such projects omit. Against random, popularity and no-decomposition baselines, using paired bootstrap tests, we find that dimensionality reduction does not improve retrieval accuracy. At  $k = 7$  the difference from plain cosine similarity on the scaled features is not statistically distinguishable ( $\Delta = -0.0031$ ,  $p = 0.427$ ); below that, reduction is significantly *worse*. What it buys instead is diversity and catalogue coverage. Separately, clustering the latent space recovers genre only partially (ARI = 0.434, NMI = 0.572): production-defined categories such as hip-hop and EDM are cleanly separable from audio features alone, while lineage-defined categories such as pop, rock and R&B are not. The taxonomy is not reproduced. It is reorganised.

## 1 Introduction

Recommendation systems for music lean heavily on two signals: genre metadata and collaborative filtering over listening histories. Both work, and both inherit assumptions. Genre is a marketing

and lineage taxonomy at least as much as an acoustic one, and collaborative filtering recommends what similar listeners chose, which entrenches whatever they were offered.

A third signal is available: the sound itself. Streaming platforms expose per-track acoustic descriptors — danceability, energy, acousticness, tempo and others — computed from the audio. If musical similarity is partly an acoustic fact rather than purely a social one, those descriptors should support recommendation on their own, and the structure they induce should tell us something about how well the genre taxonomy tracks how music actually sounds.

This paper builds such a system and, more importantly, measures it.

## Contributions.

- A recommendation pipeline built on a from-scratch SVD, with an algorithm choice justified by measured numerical accuracy rather than convention (Section 3).
- Explicit criteria for choosing the latent dimension, and evidence that the standard criteria disagree sharply on this kind of data (Section 4).
- An evaluation harness with three baselines and paired significance testing, yielding a negative result about the central method (Section 7).
- A quantitative answer to whether audio features reproduce genre, using chance-corrected agreement metrics (Section 8).

**A note on data.** Reproducing this work on the obvious corpus is harder than it looks. The widely-cited *Spotify Top Songs and Audio Features* collection [6] is not a published file: the repository contains a notebook that *builds* a dataset, requiring weekly chart exports from a separate scraper project and personal Spotify API credentials. Spotify has since restricted the `audio-features` endpoint for newly created API applications, so that corpus is not straightforwardly reconstructible.

All quantitative results below therefore come from a synthetic catalogue of 2,997 tracks with known generative structure, described in Section 2. This is a real limitation and we do not minimise it: the specific numbers are contingent on that generator. The *method* is not, and every figure and number in this paper regenerates from a single command against any catalogue with the required columns (Section 10).

## 2 Dataset and preprocessing

Each track is represented as a vector of nine acoustic descriptors: danceability, energy, speechiness, acousticness, instrumentality, liveness, valence, loudness and tempo. Figure 1 shows the format such data typically arrives in.

| id                    | artist_names                                               | track_name                           | source                                    | key    | mode    | time_signature | danceability | energy | speechiness | acousticness | instrumentalness | liveness | valence | loudness | tempo   |
|-----------------------|------------------------------------------------------------|--------------------------------------|-------------------------------------------|--------|---------|----------------|--------------|--------|-------------|--------------|------------------|----------|---------|----------|---------|
| 000G6u2NkLzdrIgrB     | ZAYN, PARTYNEXTDOOR                                        | Still Got Time (feat. PARTYNEXTDOOR) | RCA Records Label                         | G      | Major   | 4 beats        | 0.748        | 0.627  | 0.0639      | 0.131        | 0.0              | 0.0852   | 0.524   | -6.029   | 120.963 |
| 003eolwaETaUWmNfMoZy  | Alessia Cara                                               | Growing Pains                        | Def Jam Recordings                        | C#m/D# | Minor   | 4 beats        | 0.353        | 0.755  | 0.733       | 0.0822       | 0.0              | 0.39     | 0.437   | -6.276   | 191.153 |
| 003vvz7NlydyvhrvR4u6B | The Killers                                                | Mr. Brightside                       | Island Records                            | C#m/D# | Major   | 4 beats        | 0.352        | 0.911  | 0.0747      | 0.00121      | 0.0              | 0.0995   | 0.236   | -5.23    | 146.033 |
| 0087TZ0Xawar6NZ0uFumN | Cardi B, Chance the Rapper                                 | Best Life (feat. Chance The Rapper)  | Atlantic/RSR                              | A      | Major   | 4 beats        | 0.62         | 0.625  | 0.553       | 0.287        | 0.0              | 0.314    | 0.665   | -7.438   | 167.911 |
| 008lm7zeNqgYLPWwagIq  | Post Malone, The Weeknd                                    | One Right Now (with The Weeknd)      | Republic Records                          | C#m/D# | Major   | 4 beats        | 0.687        | 0.781  | 0.053       | 0.0361       | 0.0              | 0.0755   | 0.688   | -4.806   | 97.014  |
| 00EPEXKJfJfH8G0b0eod  | Thalia, NATTI NATASHA                                      | No Me Acuerdo                        | Sony Music Latin                          | G      | Minor   | 4 beats        | 0.636        | 0.799  | 0.9673      | 0.187        | 0.0              | 0.082    | 0.772   | -4.247   | 94.033  |
| 00ETwH0DopgS0WV0VWt   | Kygo, Donna Summer                                         | Hot Stuff                            | RCA Records Label                         | F      | Major   | 4 beats        | 0.681        | 0.773  | 0.146       | 0.019        | 1.286-06         | 0.11     | 0.429   | -5.749   | 119.961 |
| 0025uP8z0z0w0M0w0z    | Tropes do Bruno, DJ Ws da Igreja, SMU, Tiz, Mc Menor Thall | Baile do Bruno                       | Tropes do Bruno                           | G      | Minor   | 5 beats        | 0.734        | 0.228  | 0.53        | 0.899        | 0.0962           | 0.133    | 0.522   | -4.731   | 162.534 |
| 00gq0R0M27m0F7AFu9fKx | YBN Nahor                                                  | Bounce Out With That                 | 2018 G#m/A#                               | Major  | 4 beats | 0.857          | 0.56         | 0.173  | 0.0426      | 0.0          | 0.153            | 0.462    | -8.278  | 94.949   |         |
| 00lmgpRfYRfM0u0u03hmk | Brent Faiyaz                                               | LOOSE CHANGE                         | Lost Kids LLC, Marketed by Venice / Stern | C#m/D# | Minor   | 4 beats        | 0.574        | 0.368  | 0.9814      | 0.753        | 0.0              | 0.147    | 0.44    | -8.931   | 84.975  |
| 00k0W654j0X18E3p9Hf   | Chance the Rapper, Macklem\$0, DaBaby                      | Hot Shower                           | Chance the Rapper                         | A      | Major   | 4 beats        | 0.899        | 0.509  | 0.387       | 0.00157      | 0.0              | 0.0573   | 0.599   | -8.04    | 150.001 |

**Figure 1:** Example tracks with their audio features. Note the `id` column: chart datasets identify tracks by URI, which is not the same as identifying them by song — a distinction that matters in Section 2.1.

## 2.1 Deduplication is not optional

Chart datasets contain one row per track *per chart week*. A track that spent eleven weeks in a Top 200 appears eleven times, with eleven nearly identical feature vectors. This is a trap with real consequences for a recommender.

The first is redundancy: near-duplicate rows crowd the nearest-neighbour ranking, so a result list of ten can easily contain only three or four distinct songs. The second is subtler. A recommender must exclude the seed playlist from its own results, and the natural implementation excludes by row index. But a seed occupies many rows, so excluding the one that matched leaves the others eligible — and the system returns the user’s own input back to them at similarity 1.000.

We collapse to one row per (title, artist), aggregating features by median so that a single mis-joined row cannot move a track. On our catalogue this merged 1,353 rows.

Deduplicating by platform URI, which is the obvious key and the one upstream tooling tends to use, is *insufficient*: the same song commonly holds several URIs across singles, albums, remasters, explicit and clean edits, and regional releases. Title-and-artist is coarser but closer to the notion of “song” that a listener has in mind.

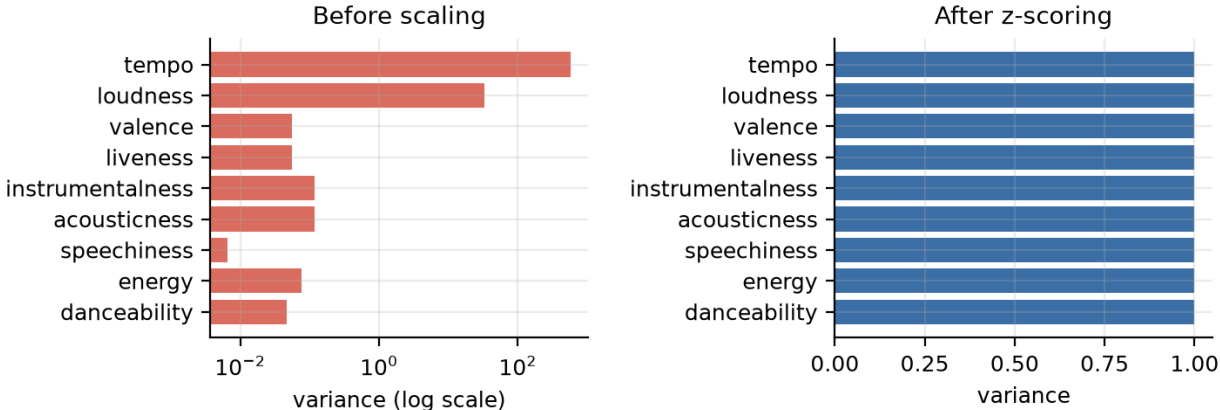
Collapsing also *creates* information rather than discarding it. The number of merged rows is a chart-persistence count, which we retain as a popularity signal and use for novelty weighting in Section 6.

## 2.2 Scaling

Features are z-scored,

$$z = \frac{x - \mu}{\sigma}, \tag{1}$$

so that each contributes comparably to distance computations. This is not a formality. Tempo is measured in the hundreds of beats per minute while danceability lies in  $[0, 1]$ ; on raw values, tempo alone accounts for 94.6% of total variance (Figure 2). Without scaling, the decomposition would be a study of tempo and essentially nothing else.



**Figure 2:** Feature variance before and after z-scoring. The left panel is on a logarithmic axis — on a linear one the other eight features are invisible.

For heavily skewed descriptors such as speechiness and instrumentalness, which are near zero for most tracks with a long tail, a robust scaler (median and interquartile range) is also available; we use z-scoring throughout for comparability.

### 3 Singular value decomposition

#### 3.1 Mathematical background

For any  $A \in \mathbb{R}^{m \times n}$ , the singular value decomposition is

$$A = U\Sigma V^T, \quad (2)$$

where  $U$  is  $m \times m$  orthogonal,  $\Sigma$  is  $m \times n$  diagonal with singular values in descending order, and  $V$  is  $n \times n$  orthogonal.

The classical derivation starts from  $A^T A$ , which is symmetric and positive semi-definite. The spectral theorem guarantees it an orthonormal eigenbasis:

$$A^T A v_i = \sigma_i^2 v_i, \quad (3)$$

and these  $v_i$  form the columns of  $V$ . For the left singular vectors, multiplying through by  $A$  gives  $Av_i = \sigma_i u_i$ , so  $u_i = Av_i / \sigma_i$  whenever  $\sigma_i \neq 0$ . If  $\text{rank}(A) = r$ , only the first  $r$  singular values are positive and

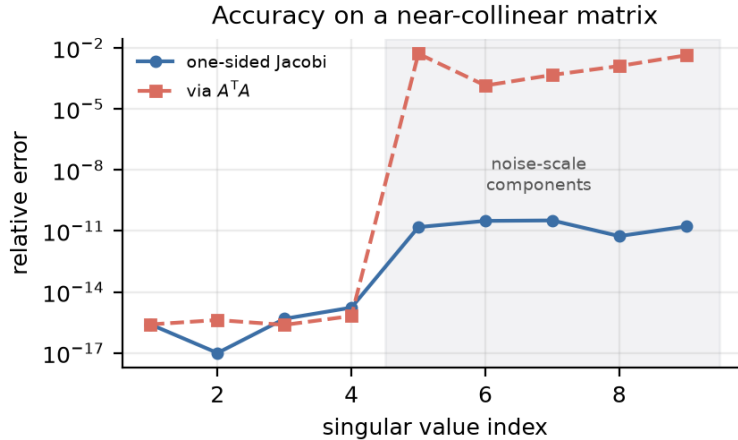
$$A = U\Sigma V^T = \sum_{i=1}^r \sigma_i u_i v_i^T, \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0. \quad (4)$$

The columns of  $V$  and  $U$  corresponding to non-zero singular values give orthonormal bases for the row space and column space of  $A$  respectively. Those orthonormal bases are what make the decomposition suitable for projection.

#### 3.2 Choosing the algorithm

The derivation above is correct mathematics and a poor numerical recipe. Forming  $A^T A$  squares the condition number: if  $\kappa(A) = \sigma_1 / \sigma_n$ , then  $\kappa(A^T A) = \kappa(A)^2$ . Singular values below  $\sqrt{\epsilon} \sigma_1$  are lost to rounding before an eigensolver ever sees them.

This is not a hypothetical concern for acoustic features, which are strongly correlated — energy with loudness, both against acousticity. Figure 3 measures the effect on a matrix with four dominant directions and five at noise scale: the  $A^T A$  route reaches a relative error of  $5.13e-03$  on the trailing components, while one-sided Jacobi holds  $3.27e-11$ .



**Figure 3:** Relative error per singular value on a near-collinear matrix. The two algorithms are indistinguishable on the leading components and differ by eight orders of magnitude on the trailing ones.

One-sided Jacobi avoids the problem by never forming  $A^\top A$ . It orthogonalises the columns of  $A$  directly through plane rotations  $A \leftarrow A J(p, q, \theta)$ , each chosen so that columns  $p$  and  $q$  become mutually orthogonal. Rotations are orthogonal transformations, so the singular values are preserved exactly at every step. At convergence  $A = U\Sigma$ , which gives

$$\sigma_i = \|a_i\|, \quad u_i = a_i/\sigma_i, \quad (5)$$

with  $V$  the accumulated product of rotations. The rotation angle for each pair comes from the  $2 \times 2$  Gram block, exactly as in the Jacobi eigenvalue algorithm, but applied one-sidedly. This is the method behind LAPACK’s `dgesvj`, and it computes singular values to high *relative* accuracy [1].

**A caution about how to measure this.** Reconstruction error does not reveal the difference. Both algorithms reconstruct  $A$  to  $\sim 10^{-16}$  relative Frobenius error, because that norm is dominated by the largest singular values — an algorithm can mangle every small component and still appear excellent. It is relative error *per singular value* that discriminates. Reaching for the obvious metric would hide the problem entirely.

### 3.3 Projection

We retain the top  $k$  right singular vectors and project:

$$B_k = B \cdot V_k. \quad (6)$$

It may seem odd to call a matrix multiplication a projection, but for an orthonormal basis the two coincide. Projecting a vector  $\mathbf{v}$  onto  $\{\mathbf{w}_1, \dots, \mathbf{w}_k\}$  gives

$$\text{proj}(\mathbf{v}) = \sum_{i=1}^k (\mathbf{v} \cdot \mathbf{w}_i) \mathbf{w}_i, \quad (7)$$

and collecting the  $\mathbf{w}_i$  as columns of a matrix  $W$ , the coordinates in that basis are

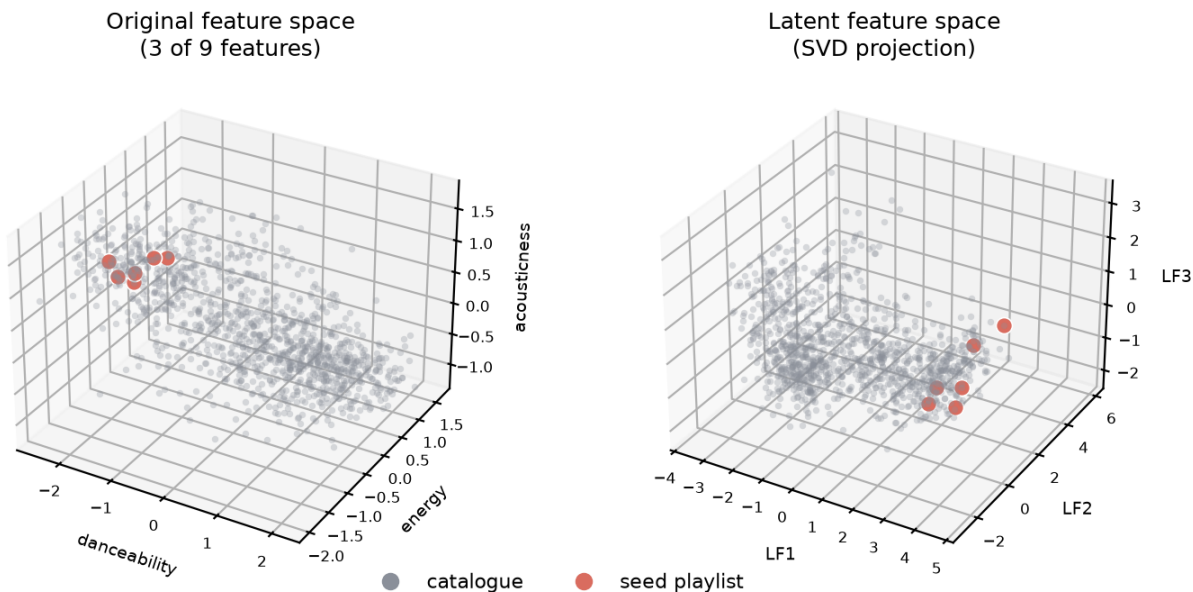
$$\mathbf{v}^\top W = [(\mathbf{v} \cdot \mathbf{w}_1), \dots, (\mathbf{v} \cdot \mathbf{w}_k)]. \quad (8)$$

Applied to every row of  $B$  at once, this is exactly  $B_k = BV_k$ .

**An identity worth stating.** When  $k = n$ ,  $V_k$  is a full orthogonal matrix, and orthogonal transformations preserve both inner products and norms. Cosine similarity is therefore *invariant* under the untruncated projection: an SVD-based recommender at full rank is mathematically identical to one using no decomposition at all.

Two things follow. First, this is a strong end-to-end correctness check, and we use it as one in Section 7 — any error in scaling, decomposition or ranking would break the equality. Second, it sharpens the research question. Whatever the decomposition contributes, it contributes entirely through *truncation*. The interesting quantity is not whether to use an SVD but how much to throw away.

### 3.4 Geometric visualisation

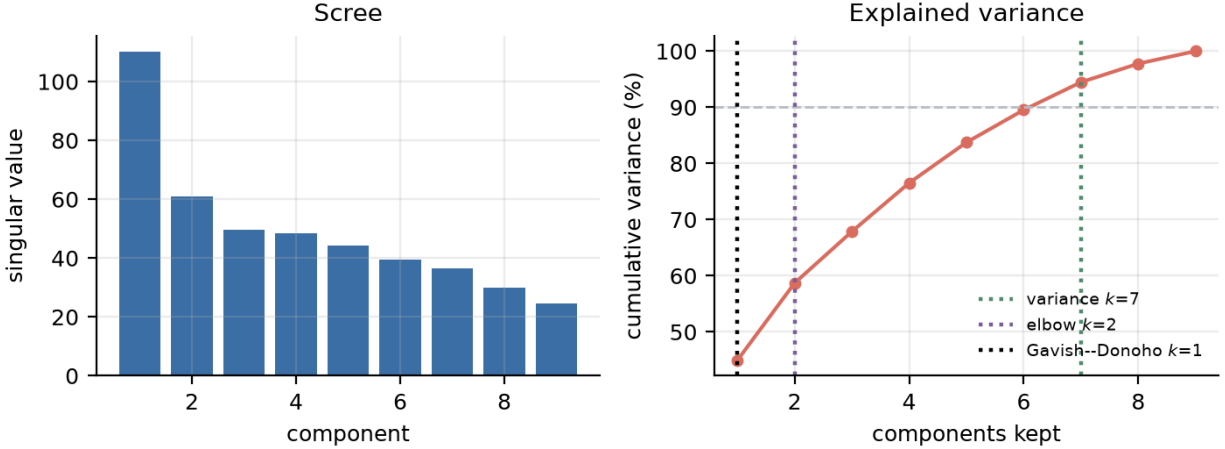


**Figure 4:** A seed playlist (coral) against the catalogue (grey), shown in three raw features and in the leading three latent dimensions. The latent view concentrates the playlist, which is what makes proximity in this space a usable ranking signal.

## 4 Choosing the number of dimensions

The parameter  $k$  controls how much information the recommender is permitted to see, so it deserves an argument rather than a convention. We implement three standard criteria:

- **Variance threshold:** the smallest  $k$  whose cumulative explained variance reaches 90%. Interpretable, but the threshold is arbitrary.
- **Elbow:** the knee of the scree curve, located by maximum perpendicular distance from the chord joining its endpoints. No threshold to choose, but it presumes a knee exists.
- **Gavish–Donoho:** the asymptotically optimal hard threshold for recovering a low-rank signal from i.i.d. Gaussian noise, which requires no free parameter at all — the matrix aspect ratio determines everything [2].



**Figure 5:** Scree plot and cumulative explained variance, with each criterion’s choice marked. They disagree sharply:  $k = 7$ ,  $k = 2$  and  $k = 1$  respectively.

That disagreement is itself a finding, and it is not a defect in any of the three rules — they are answering different questions. Gavish–Donoho asks how many components exceed what noise alone would produce, and answers 1. The variance criterion asks how many are needed to reconstruct the data faithfully, and answers 7. When a spectrum has one dominant component followed by a near-level tail, as ours does (Figure 5, left), those questions have very different answers, and a flat tail is exactly the regime in which retained components are hard to distinguish from noise. This anticipates the result in Section 7.

**Reporting explained variance.** One arithmetic detail is worth stating because it is easy to get wrong. Explained variance must be denominated against the *full* spectrum:

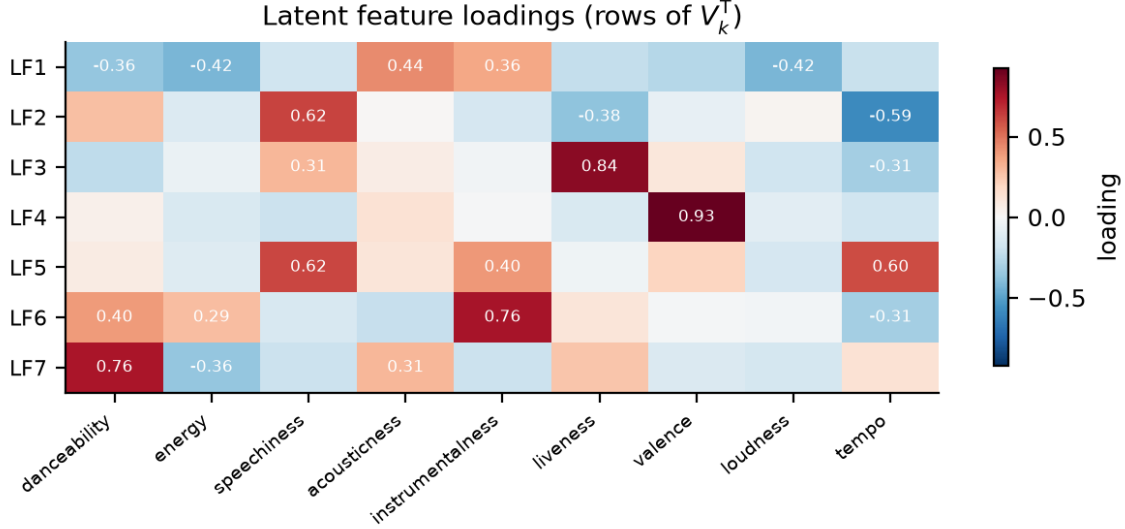
$$\text{EVR}_i = \frac{\sigma_i^2}{\sum_{j=1}^n \sigma_j^2}, \quad (9)$$

where the sum runs over all  $n$  components, not merely the  $k$  retained. Normalising a truncated spectrum against itself makes every configuration appear to explain 100% of variance regardless of  $k$ , which is both false and conveniently flattering. Our model retains 94.5% across 7 of 9 components.

## 5 Latent features and explanation

Figure 6 shows the loadings — each latent component expressed as a weighting of the original descriptors. Because the components are mutually orthogonal, each captures an uncorrelated axis of acoustic variation.

Signs are canonicalised: the largest-magnitude entry of each right singular vector is forced positive. This matters for interpretation. The decomposition is unique only up to a simultaneous sign flip of  $u_i$  and  $v_i$ , so without pinning a convention a component described in prose as “high energy, loud” can reappear as “low energy, quiet” on a different machine or a different slice of data, making any written interpretation unreliable.



**Figure 6:** Latent feature loadings, the rows of  $V_k^T$ . Each row is one component expressed as a weighting of the original audio features.

Orthonormality also gives us explanation for free. For unit-normalised latent vectors  $\hat{q}$  and  $\hat{r}$ , cosine similarity decomposes exactly:

$$\cos(q, r) = \sum_{i=1}^k \hat{q}_i \hat{r}_i, \quad (10)$$

so the term  $\hat{q}_i \hat{r}_i$  is precisely the contribution of component  $i$  to the match, and the terms sum to the score. Paired with that component’s loadings, this turns every recommendation into a sentence: *matched on the energy–loudness axis, which both tracks score high on*. A recommender that reports only a similarity score asks the reader to take the mathematics on faith; the decomposition needed to avoid that is already computed.

## 6 Recommendation strategies

We implement four strategies over the latent representation.

- **One-per-song** — one recommendation per seed track, ensuring every input contributes and maximising coverage of the playlist.
- **Overall top** — the globally highest-scoring candidates, maximising coherence with whatever dominates the input.
- **Centroid** — query by the playlist’s mean latent vector, its “taste vector”. This differs from averaging per-seed scores: it is more sensitive to a playlist that is coherent in direction and less to a single outlying seed.
- **Maximal Marginal Relevance** — greedily select the candidate maximising

$$\lambda \cdot \text{rel}(d) - (1 - \lambda) \max_{s \in S} \text{sim}(d, s),$$

so each pick must be both close to the playlist and unlike what has already been chosen [4].



The first two sit at opposite ends of a relevance–diversity spectrum with nothing in between: one maximises similarity, the other enforces variety by construction. MMR makes that trade-off an explicit parameter rather than a choice of algorithm, and  $\lambda$  can be tuned against the metrics in Section 7.

Per-seed scores are aggregated by `max`, `mean` or Borda count. The choice is not cosmetic: under `max`, a single atypical seed can dominate the entire result list, a real hazard for playlists containing outliers or duplicated entries. Borda count aggregates ranks rather than scores and is immune to one seed having systematically higher cosine values than another.

We additionally support a per-artist cap, popularity de-biasing (subtracting a weighted popularity term so that results are not simply the chart re-ordered), and Rocchio-style negative feedback that steers the query away from tracks the listener rejects.

## 7 Evaluation

### 7.1 Protocol

There is no ground truth for enjoyment, so we borrow one: tracks sharing an artist are treated as mutually relevant. For each of 219 evaluation groups we sample three tracks as a query playlist and hold out the artist’s remaining tracks as the relevant set. Seeds are excluded from results, so no system can score by returning the query.

We report accuracy (hit rate, recall, MRR, NDCG) alongside beyond-accuracy measures (intra-list diversity, catalogue coverage). Both families are necessary: a system returning the ten most popular songs for every query can score respectably on accuracy while being useless, and only coverage and diversity expose that. Diversity is computed in the shared scaled-feature space rather than in any model’s own latent space, so that no system is graded in the geometry most flattering to it.

**What this protocol is not.** It measures whether a system recovers known stylistic neighbours from audio alone. It structurally *penalises* the cross-genre discovery that motivates the project: a genuinely surprising good recommendation lies outside the ground-truth group and scores as a miss. The numbers should be read as “does the latent space encode real structure”, not as “is this a good product”.

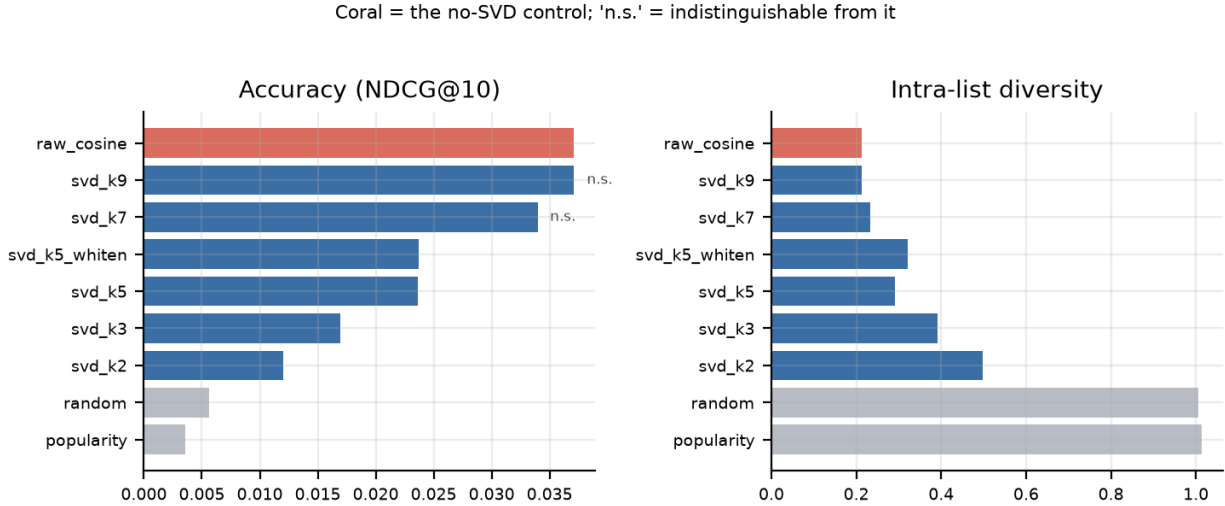
### 7.2 Baselines

- **Random** — the floor. Any metric a system cannot beat here is measuring noise.
- **Popularity** — ignores the query entirely and returns the most popular tracks. Deceptively strong on accuracy metrics, which is exactly why it belongs.
- **Raw cosine** — cosine similarity on the nine scaled features, with no decomposition. **This is the control that matters.** The entire justification for the SVD is that projecting to a latent subspace beats using the features directly, and that is a hypothesis rather than a premise.

### 7.3 Results

| System              | HR@10  | R@10   | MRR    | NDCG@10 | Div.   | Cov.   |
|---------------------|--------|--------|--------|---------|--------|--------|
| <b>raw_cosine</b>   | 0.3242 | 0.0351 | 0.0865 | 0.0371  | 0.2125 | 0.4698 |
| svd_k9 <sup>†</sup> | 0.3242 | 0.0351 | 0.0865 | 0.0371  | 0.2125 | 0.4698 |
| svd_k7 <sup>†</sup> | 0.3059 | 0.0322 | 0.0793 | 0.0340  | 0.2331 | 0.4875 |
| svd_k5_whiten       | 0.2009 | 0.0224 | 0.0559 | 0.0237  | 0.3208 | 0.5032 |
| svd_k5              | 0.2283 | 0.0241 | 0.0508 | 0.0236  | 0.2903 | 0.5052 |
| svd_k3              | 0.1370 | 0.0153 | 0.0412 | 0.0169  | 0.3904 | 0.5128 |
| svd_k2              | 0.0959 | 0.0086 | 0.0338 | 0.0120  | 0.4973 | 0.5189 |
| random              | 0.0457 | 0.0057 | 0.0154 | 0.0056  | 1.0046 | 0.5202 |
| popularity          | 0.0411 | 0.0033 | 0.0088 | 0.0036  | 1.0131 | 0.0037 |

**Table 1:** Held-out-artist evaluation over 219 queries. Higher is better throughout. <sup>†</sup> marks systems whose NDCG difference from **raw\_cosine** is not statistically significant under a paired bootstrap test.



**Figure 7:** Accuracy and diversity. Every latent model beats random and popularity by a wide margin — the features carry real signal. None beats the no-decomposition control on accuracy.

Three findings follow from Table 1.

**The pipeline is correct.** *svd\_k9* reproduces *raw\_cosine* to every reported digit on every metric ( $\Delta = 0$ ). This is the identity predicted in Section 3: at full rank the projection is an orthogonal rotation, and cosine similarity is rotation-invariant. It is a demanding end-to-end check, since an error anywhere in scaling, decomposition or ranking would break the equality.

**Truncation to  $k = 7$  costs nothing measurable.** *svd\_k7* scores 0.0340 against 0.0371 for raw cosine. A paired bootstrap over the shared queries gives  $\Delta = -0.0031$  with a 95% confidence interval of  $[-0.0105, +0.0048]$  and  $p = 0.427$ . The interval contains zero: on this evidence the two are indistinguishable. Meanwhile intra-list diversity rises from 0.2125 to 0.2331 and catalogue coverage from 0.4698 to 0.4875. Discarding two dimensions is free, and it broadens the results.

**Aggressive truncation is significantly worse.** For  $k \leq 5$  every comparison against raw cosine is significant at  $p < 0.01$ , and accuracy falls monotonically as  $k$  decreases.

**On pairing.** All systems are evaluated on identical queries, so the per-query differences share the query-difficulty variance. Testing them as independent samples would discard that shared structure, overstate the uncertainty, and risk converting a null result into a spurious one — or the reverse. The comparison must be paired, and the bootstrap must resample query indices rather than each system separately.

## 7.4 Interpretation

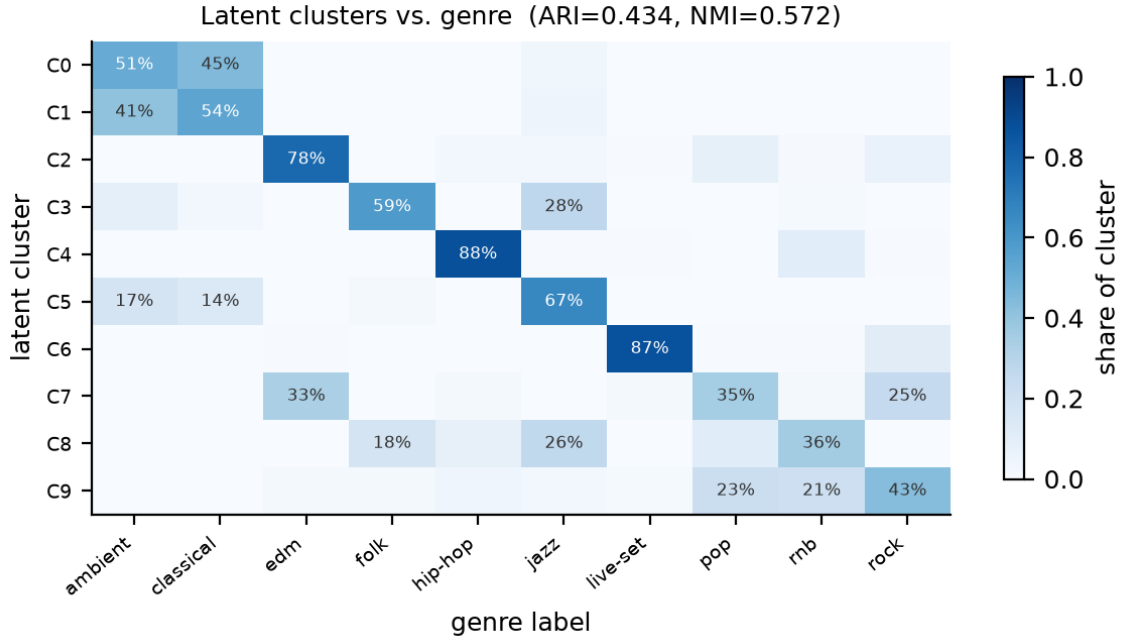
The honest summary is that dimensionality reduction is not an accuracy technique in this setting. It is a diversity technique. There is a regime ( $k = 7$ ) where it is free, and below that one trades measurable accuracy for measurable variety. That may well be the right trade for a discovery product, where surfacing unfamiliar material is the point — but it should be made deliberately rather than assumed, and the shape of the trade is now something this system can report rather than something its author has to claim.

It is worth stating plainly what this does *not* show. It does not show the SVD is useless: the full-rank identity means the decomposition is exactly as good as the raw features by construction, and at  $k = 7$  the representation carries two fewer dimensions than the original 9 with no measurable loss, which matters at catalogue scale. Nor does it show that latent structure is absent — Section 8 demonstrates the opposite. It shows that on this retrieval task, compression does not buy accuracy.

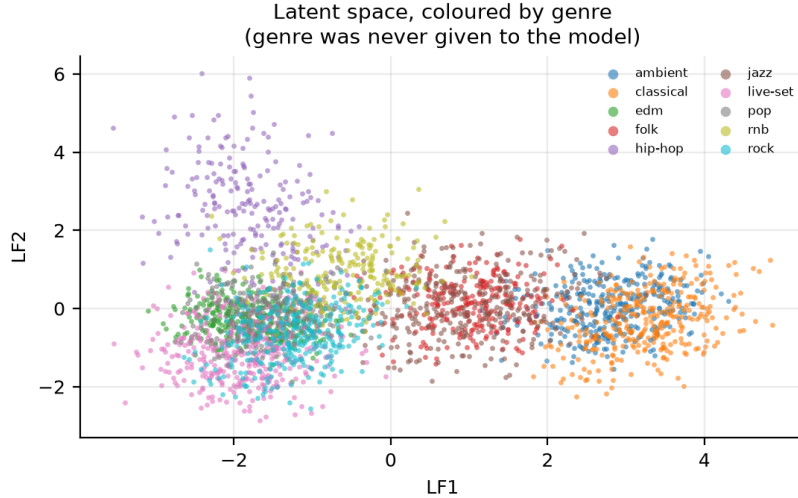
## 8 Does audio-feature classification reproduce genre?

We now return to the question posed in the abstract. Recommendation quality cannot answer it; clustering can.

We partition the latent space with  $k$ -means (k-means++ seeding, ten restarts) into as many clusters as there are genre labels, then measure agreement between the partition and the labels. The metrics are chosen to be invariant to label permutation — cluster 3 has no intrinsic relationship to “jazz”, so any measure depending on which integer was assigned to which group is meaningless here — and corrected for chance: Adjusted Rand Index [5] and Normalised Mutual Information. Purity is reported alongside but not relied upon, since it can be driven to 1 simply by using more clusters.



**Figure 8:** Each latent cluster’s composition by genre label. Rows spanning several columns are genres the audio features consider interchangeable; columns spread across rows are genres the features consider internally inconsistent.



**Figure 9:** The catalogue in its two leading latent dimensions, coloured by a genre label the model never received.

Agreement is partial:  $ARI = 0.434$ ,  $NMI = 0.572$ ,  $purity = 0.591$  and  $silhouette = 0.178$  across 10 genres. The partition partially reproduces the reference taxonomy. Neither extreme holds — the taxonomy is neither recovered nor unrelated — and the structure visible in Figure 8 is considerably more informative than the scalar summaries:

- **Cleanly separable.** Hip-hop (88% of its cluster), live recordings (87%) and EDM (78%) each occupy a cluster of their own. These are categories defined by *production*: speechiness,

audience noise, synthetic timbre. Audio descriptors measure production directly.

- **Merged.** Ambient and classical are split across two clusters, each of which is a mixture of both. The features cannot separate them at all — both are quiet, acoustic, instrumental and low-energy — even though the two are historically and culturally distinct.
- **Dissolved.** Pop, rock and R&B form no clusters of their own, spreading instead across three mixed groups. These are categories defined by lineage, instrumentation convention and marketing, none of which the nine descriptors observe.

The answer, then, is: *reorganised, but not arbitrarily*. Audio features impose an axis of production character rather than genre ancestry. Where a genre label happens to encode production, it survives the translation; where it encodes history, it dissolves. This is a coherent alternative taxonomy rather than noise, and it suggests that the disagreements between acoustic similarity and genre metadata are systematic and predictable rather than incidental.

**Caveat.** This result is measured on a synthetic catalogue whose genre profiles were specified by hand, and the observed merges partly reflect overlaps in those specifications — ambient and classical were given deliberately similar acoustic profiles. The methodology transfers directly; this particular finding should be re-measured on a real corpus before being relied upon, and doing so requires one command (Section 10).

## 9 Implementation notes

Every decomposition is implemented from the underlying mathematics: Householder QR, two symmetric eigensolvers (shifted QR with deflation, and cyclic Jacobi), one-sided Jacobi SVD, randomized SVD [3], cosine similarity, Levenshtein distance for track lookup,  $k$ -means, and all ranking and agreement metrics. NumPy supplies array storage and elementwise operations; `numpy.linalg` appears only in the test suite, as ground truth.

Three implementation details proved to matter more than expected, and are worth recording for anyone building something similar.

**Convergence tolerances must be relative.** An iterative eigensolver that tests convergence by comparing an absolute off-diagonal sum against a fixed constant will behave differently at different catalogue sizes, because that sum scales with  $\|A\|$ . On a large matrix the tolerance becomes unreachable and the solver silently exhausts its iteration budget every time. Both of our eigensolvers scale their tolerance by the Frobenius norm.

**Sign conventions must be pinned.** As noted in Section 5, an unconstrained SVD produces components whose sign is an artefact of the arithmetic, which makes any prose interpretation of a loading unreliable across runs.

**Terminal output must survive the local encoding.** Catalogue metadata is full of accented and non-Latin track titles, and a console using a legacy code page will raise an encoding error rather than print them. All output degrades to ASCII when the active encoding cannot represent a character, and continuous integration runs the test suite under a legacy code page as well as UTF-8.

The system is verified by 326 tests covering reconstruction accuracy, orthogonality, Eckart–Young optimality of the rank- $k$  truncation, cross-backend agreement, and the behaviour of each metric at its known extremes.

## 10 Reproducibility

Every figure, table and quoted number in this paper is produced by a single command, so the document cannot drift from the code that produced it:

```
python paper/build.py
python paper/build.py --data data/songs.csv    % any catalogue
```

The build regenerates each figure, writes the results table, and emits every quoted quantity as a LaTeX macro that the prose references in place of a literal. Changing  $k$ , the scaling, or the ranking strategy and rebuilding updates the text along with the figures.

## 11 Limitations

- **Synthetic corpus.** All results come from a generated catalogue (see §1). Its generative structure is known and influences the outcome, particularly the genre merges of Section 8.
- **Proxy ground truth.** Shared authorship stands in for relevance. It penalises exactly the cross-genre discovery this system aims at, and conflates an artist’s stylistic consistency with acoustic similarity.
- **Nine descriptors.** The features describe production, not melody, harmony, lyrics or song structure. The dissolution of pop, rock and R&B in Section 8 may say more about that limitation than about the genres themselves.
- **Linearity.** SVD finds the best *linear* subspace. If acoustic similarity is governed by a curved manifold, a linear projection will miss it, and a nonlinear method might reach a different conclusion about the value of reduction.
- **No listener study.** Every claim here is offline. Whether any of it corresponds to what a person would enjoy is untested.

## 12 Conclusion

We set out to recommend music from acoustic descriptors rather than genre tags, and to find out whether the resulting organisation of music resembles the taxonomy we already use.

On the first question the answer is more qualified than we expected. Dimensionality reduction did not improve retrieval accuracy over the raw scaled features. At  $k = 7$  it was free; below that it cost accuracy in exchange for diversity and coverage. The full-rank identity between the SVD recommender and plain cosine similarity makes the reason precise: the decomposition contributes nothing except through truncation, and truncation is a trade rather than a gain. Reporting this required baselines and paired significance tests, and it is the kind of result a project without them would not have found.

On the second question the answer is more interesting. Audio features neither reproduce the genre taxonomy nor scramble it. They reorganise music by production character — separating

hip-hop, EDM and live recordings cleanly, merging ambient with classical, and dissolving pop, rock and R&B entirely. That reorganisation is a coherent alternative view of a catalogue rather than a failure to recover the familiar one, and it is the more promising direction for future work: not better retrieval of what genre already tells us, but description of the structure genre leaves out.

## References

- [1] J. Demmel and K. Veselić. Jacobi’s method is more accurate than QR. *SIAM Journal on Matrix Analysis and Applications*, 13(4):1204–1245, 1992.
- [2] M. Gavish and D. L. Donoho. The optimal hard threshold for singular values is  $4/\sqrt{3}$ . *IEEE Transactions on Information Theory*, 60(8):5040–5053, 2014.
- [3] N. Halko, P.-G. Martinsson, and J. A. Tropp. Finding structure with randomness: probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review*, 53(2):217–288, 2011.
- [4] J. Carbonell and J. Goldstein. The use of MMR, diversity-based reranking for reordering documents and producing summaries. In *SIGIR ’98*, pages 335–336, 1998.
- [5] L. Hubert and P. Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985.
- [6] J. Orlandi. Spotify Top Songs and Audio Features. [https://github.com/JulianoOrlandi/Spotify\\_Top\\_Songs\\_and\\_Audio\\_Features](https://github.com/JulianoOrlandi/Spotify_Top_Songs_and_Audio_Features), 2024.
- [7] G. B. Kim. SVD mathematical definitions. Stanford University, Department of Statistics.